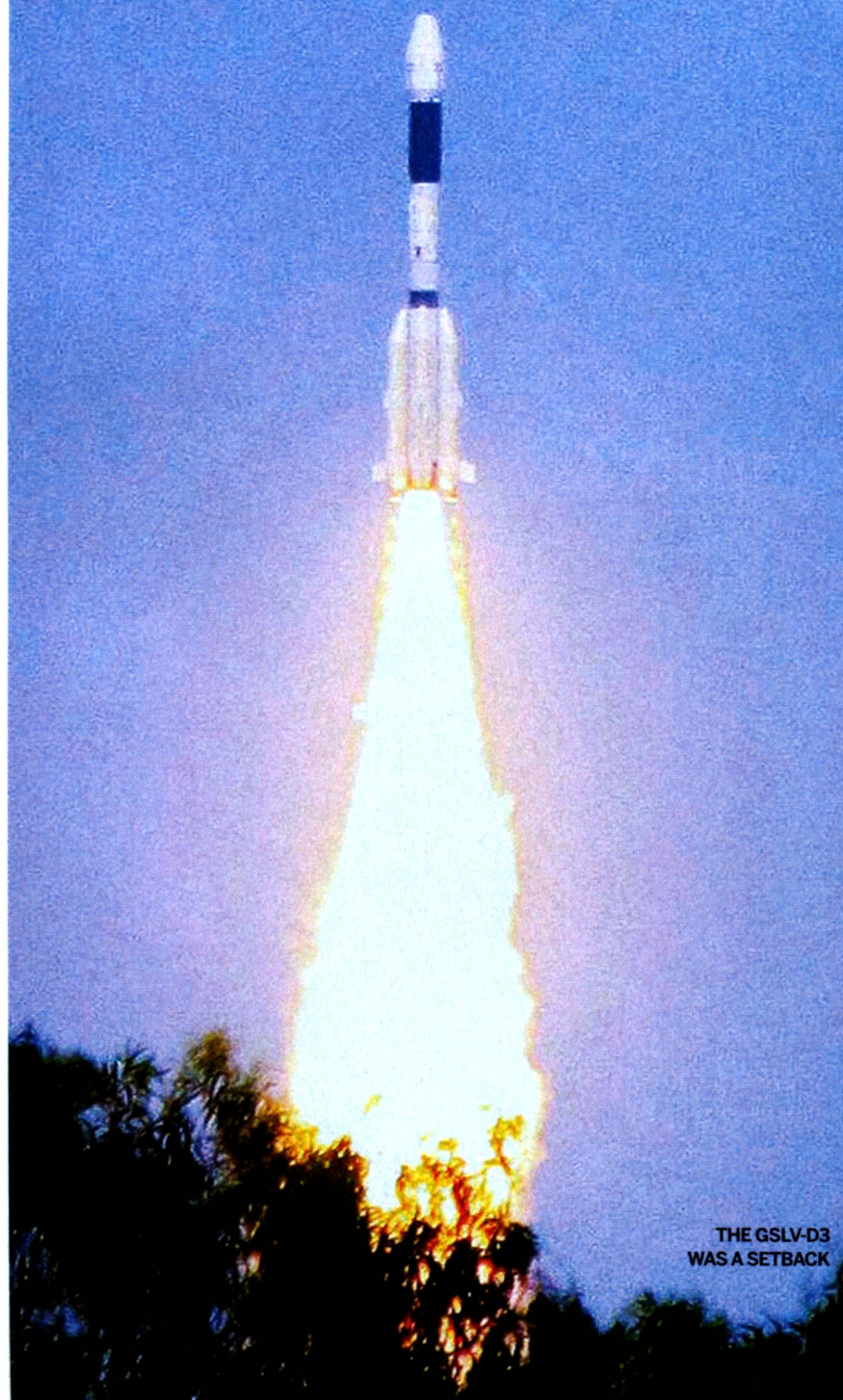


Off to a COLD START



THE GSLV-D3
WAS A SETBACK

What the failure of the indigenous cryogenic engine means for India's space ambitions and its future plans

| By Stephen David |

For nearly two decades, rocket scientists have toiled hard at the Liquid Propulsion System Centre (LPSC) on the semi-arid slopes of a 1,600-metre high mountain on the leeward side of the Western Ghats in Tamil Nadu to indigenously develop cryogenic engine technology. But their dreams soured at Sriharikota on April 15 when the 416-tonne, 17-storey high Geosynchronous Satellite Launch Vehicle (GSLV-D3), after a textbook perfect lift off failed to keep the planned trajectory as the cryogenic engine and associated smaller engines underperformed and it tumbled and did not lob the advanced communications satellite, GSAT-4, into the geostationary orbit 36,000 km above the Earth. It was a flawless flight for the first 505 seconds of the 1,022-second-long planned journey when the GSLV veered off course leading to the failure of the mission. Entry into the exclusive club of five countries that makes cryogenic engines indigenously eludes India for at least another year.

Clearly, it is a setback for ISRO as the Rs 330-crore GSLV-D3 mission, including the satellite, is a major loss. Mastering cryogenic technology is complex and all the members of the select club, except France, did not succeed with indigenous cryogenic engines on the inaugural launch. However, it is not viewed as a consolation by ISRO engineers including the GSLV-D3 Mission Director Mohammed Muslim. They spent nearly eight months on the final preparation instead of the customary two months but finally faced disappointment. "We have worked hard for the past 18 years and will be ready for the next launch with the home-made cryogenic engine in less than a year," says a crestfallen ISRO

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